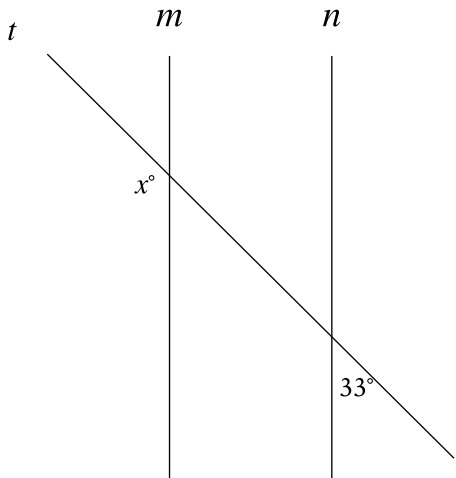


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line *m* is parallel to line *n*, and line *t* intersects both lines. What is the value of *x*?

Answer

- A. 33
- B. 57
- C. 123
- D. 147

Correct Answer: D

Rationale

Choice D is correct. It’s given that line *m* is parallel to line *n*, and line *t* intersects both lines. It follows that line *t* is a transversal. When two lines are parallel and intersected by a transversal, exterior angles on the same side of the transversal are supplementary. Thus, $x + 33 = 180$. Subtracting 33 from both sides of this equation yields $x = 147$. Therefore, the value of *x* is 147.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.